LESSON PLAN

SFNA (MOD AIRMANSHIP) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Various properties of FOLG's viscosity, absolute kinematics and dynamic viscosity, viscosity index, extreme pressure lubricants, film lubric
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Various properties of FOLG’s viscosity, absolute kinematics and dynamic viscosity, viscosity index, extreme pressure lubricants, film lubrication and boundary lubrication, determination of viscosity and viscosity index, activities in oils, service ratings of oil, SAE number
5. Specific Objectives	24.1.2.1 Trainee will be able to describe the use of lubrication, types of lubrication, viscosity and additives.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Viscosity Index determination method; Safety: Handle oil samples with gloves; Supervision: Instructor monitors handling of samples
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of lubrication in an engine?
(b) Name two types of lubrication based on film thickness.
(c) Define viscosity in simple terms.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Absolute and dynamic viscosity definitions • Viscosity index and its significance • Extreme pressure lubricants characteristics • Film lubrication and boundary lubrication types • Determination of viscosity and viscosity index	L & Ds	PPT, IB	15 min
(b)	• Activities in oils and their effects • Service ratings of oil specifications • Understanding SAE number grading system • Role of additives in oil performance • Practical demonstration of viscosity concepts	L & Ds	PPT, PDF	20 min
(c)	• Summary of lubrication properties • Review of viscosity measurement techniques • Application of SAE numbers in selection • Importance of extreme pressure lubricants • Final Q&A on boundary lubrication	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between absolute and dynamic viscosity.
(b) How does the viscosity index affect oil performance in varying temperatures?
(c) What is the purpose of extreme pressure additives in lubricants?

12. Home Assignment and Follow Up	Prepare a brief note on the differences between boundary lubrication and film lubrication, referencing the SAE number system for oil selection.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Type of grease and lubricants and their uses	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min